# Modeling Perception in Sampling for Visualizations

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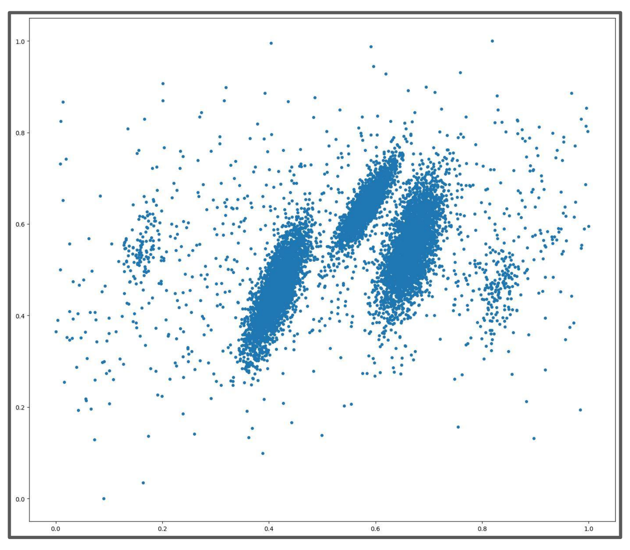
## Introduction & Motivation

- Rendering massive datasets creates **visual clutter** and is **computationally expensive**.
- We need **representative samples** that preserve important visual characteristics (e.g., clusters, outliers, patterns, spread, etc.) at low computational cost.

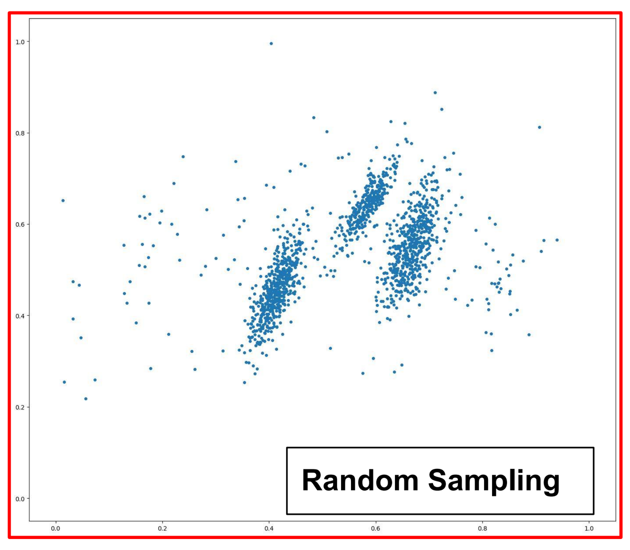


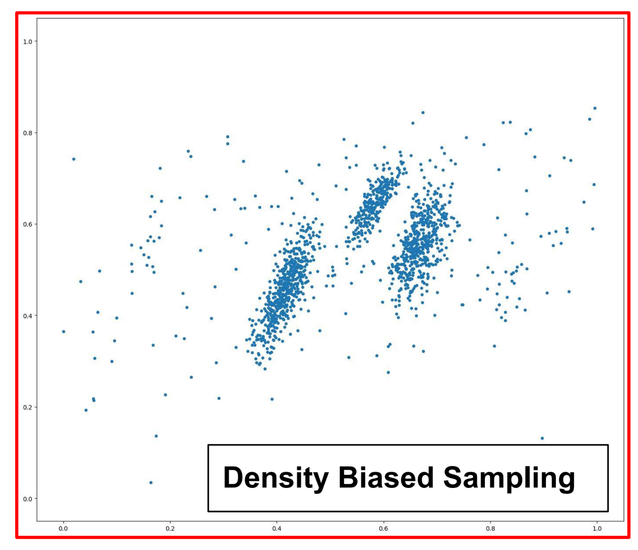
Scatterplot of the complete data

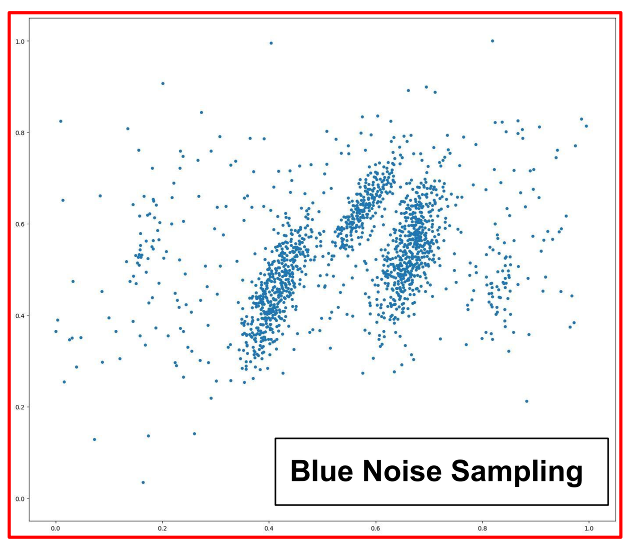
Samples

## Research Gap

Standard sampling algorithms [3] fail to represent all important visual characteristics



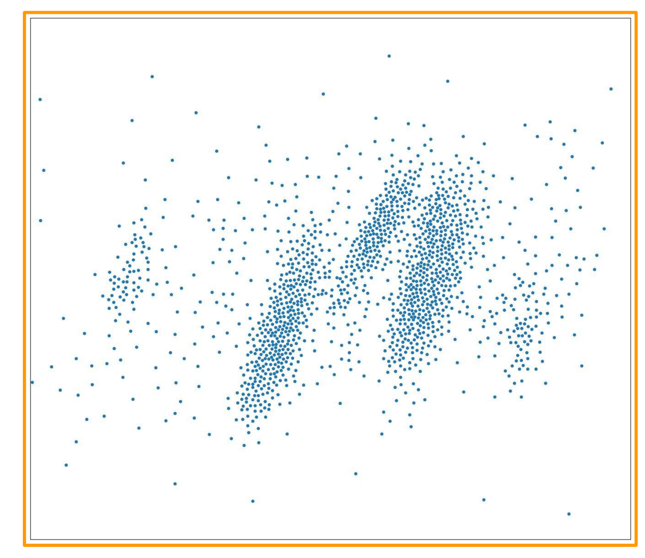




- **Smaller two clusters are either absent or not properly visible** in the samples produced by traditional sampling algorithms.
- **Data spread pattern is also not properly representative** of the original plot

## State-of-the-art: PAwS

- **Perception Aware Sampling Algorithm** (PAwS) produces samples that better represent characteristics of the original plot [1]. However it hits a **computational bottleneck**.



**PAwS**: Captures all the prevailing patterns, clusters and trends.

## Perception Aware Sampling Algorithm: **PAwS**

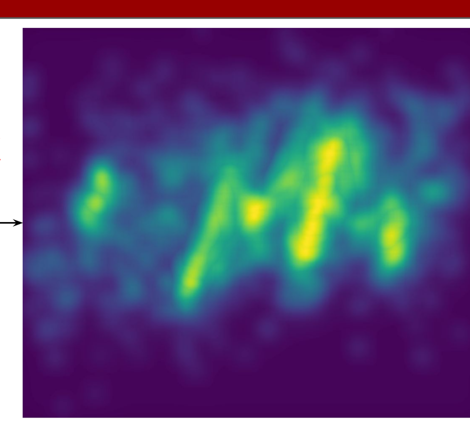
- PAwS uses **saliency models** as a proxy for **human perception**. It combines it with **density** information to produce a **perception score**.


Human eyes focus on

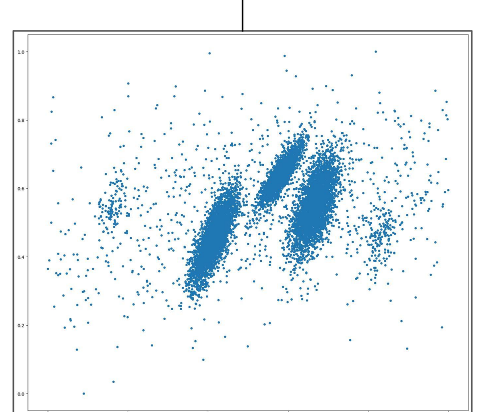
Within a plot

**Perception score** is a novel metric that models the perceptual importance of data points in a scatterplot visualization.

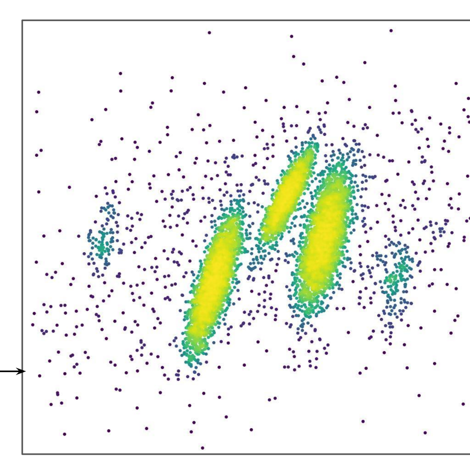
**Computational bottleneck**

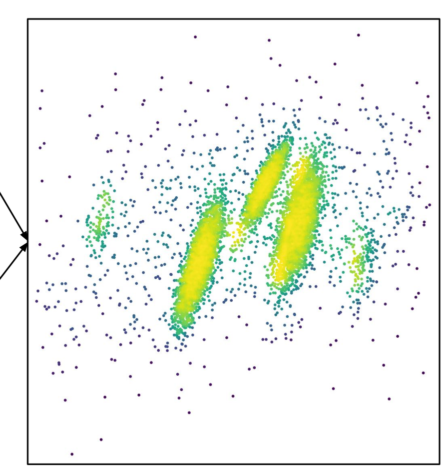


**Saliency** information, as a proxy, to model eye tracking pattern [2]



Complete data





Approximated **density** scores to locate dense regions

Used to produce sample

**Perception score** helps to quantify regions of human interest (***Higher score is denoted by brighter color*)

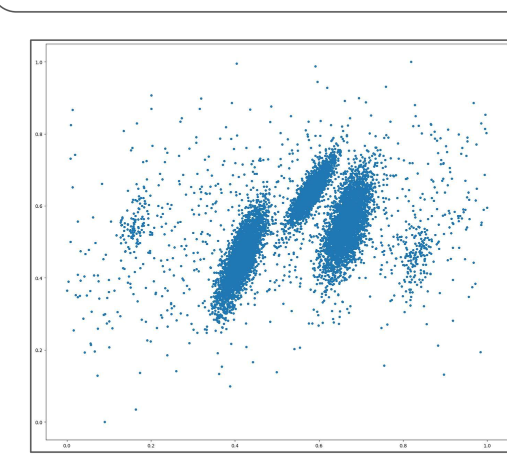
Computing saliency information requires **rendering the plot from the complete data** and running **expensive saliency models.**

## Our Contribution: Coverage-Density Score (CD Score)

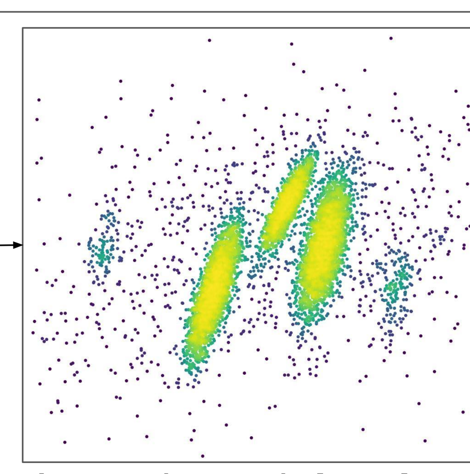
### Our Objective

- Perception score balances coverage (which better represents outliers and shapes) and density (which better represents trends and patterns).
- We eliminate the expensive saliency computation, with **CD score**, a novel **re-weighing strategy** based only on density estimation.
- PAwS adapted to use CD score achieves samples with similar visual fidelity as the original PAwS, at much faster runtimes.

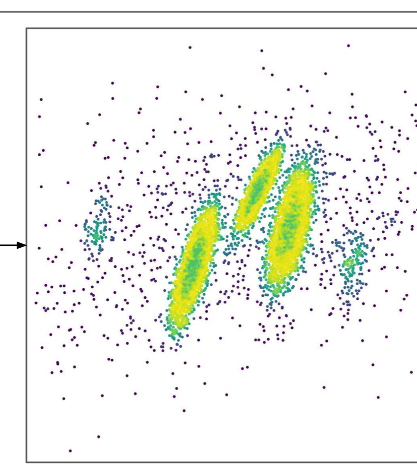
- CD Score **redistributes density scores** from **denser to sparser regions** to balance the perceptual importance of different regions of data space.



Complete data



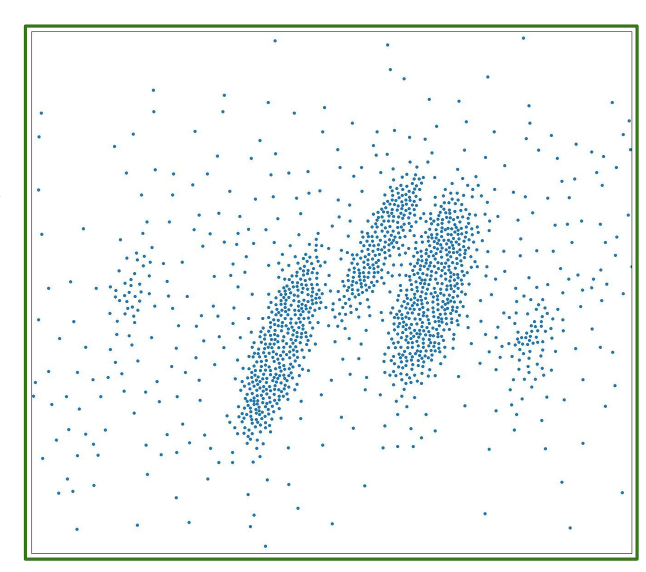
Approximated **density** scores



CD score computed by redistributing density scores

PAwS with CD score

Proposed strategy to model perception

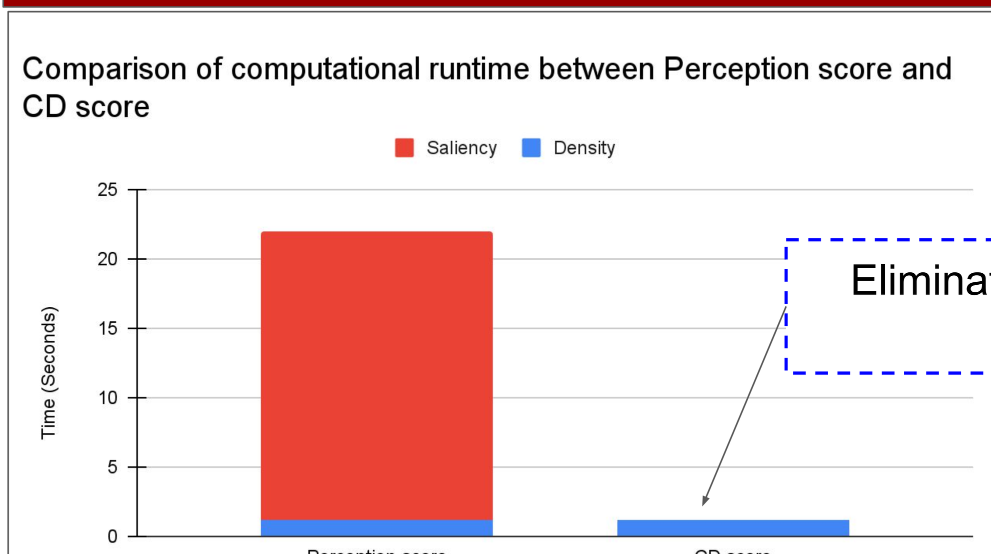


The re-weighing strategy in CD score increases the score of sparser regions, and reduces the score of denser ones.

Ensures a more balanced representation of patterns, shapes, and trends across all areas of the original plot

**PAwS sample based on CD score**: preserves all the important visual characteristics with 90% reduction in preprocessing time.

## Runtime Improvement




Elimination of saliency cost reduces preprocessing costs

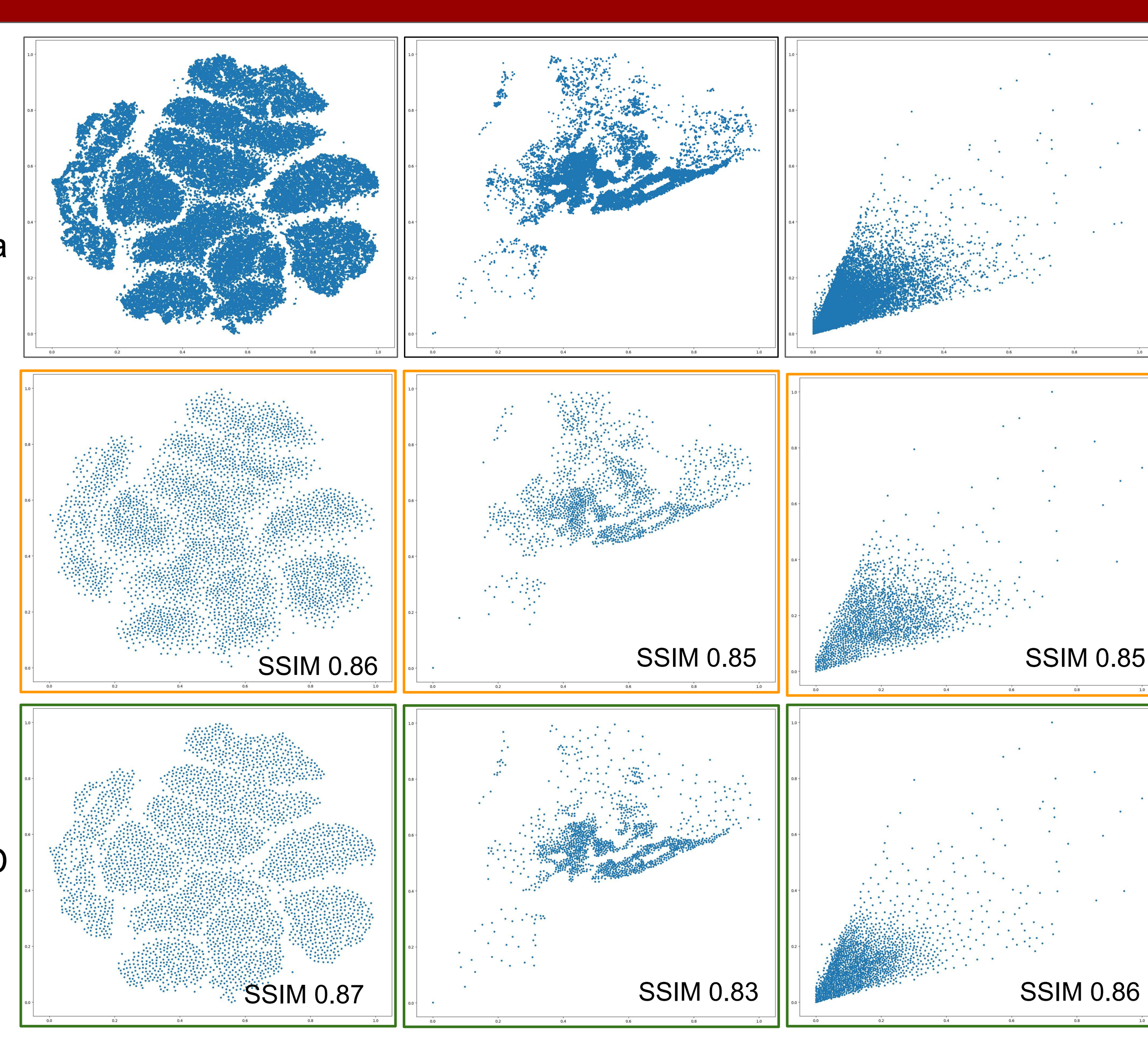
## Samples Observed in Other Datasets

**SSIM score** computes the image-based structural similarity of each sample plot with respect to original plot [4]. A higher score indicates higher similarity.

Original data

PAwS sample generated with saliency-based perception score

PAwS sample generated with CD score




## References

[1] Moumoulidou et al., "*Perception-aware Sampling for Scatterplot Visualizations*", arXiv, 2025.
[2] Matzen et al., "*Data visualization saliency model*", IEEE TVCG, 2017.
[3] Yuan et al., "*Evaluation of Sampling Methods for Scatterplots*", IEEE TVCG, 2021.
[4] Z. Wang et al., "*Image quality assessment: from error visibility to structural similarity*", IEEE TIP., 2004.